Improving Trip-Purpose Inference from GPS Trajectories using Semantically-Enriched Hidden Markov Models

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***Abstract*—*Trip-purpose inference—determining the reason behind a trip from GPS trajectory data—is a challenging problem in urban mobility research. While prior work has largely focused on transport-mode detection, the related question of why someone travels (commute, education, shopping, leisure) has received less attention, particularly using Hidden Markov Models (HMMs) augmented with geographic context. This paper investigates whether enriching HMMs with POI-based semantic features derived from a geographically grounded Beijing landmark dataset significantly improves trip-purpose classification over a trajectory-only baseline. Using the full Microsoft GeoLife dataset (182 users, 28,608 trips), we extract twelve trajectory features—including one-hot encoded time-of-day bins—and four semantic POI-count features within 300 m of each trip destination. Trip-purpose labels are generated automatically via weak supervision across five categories. Under repeated stratified 5-fold cross-validation (3×5 = 15 folds), the semantically enriched HMM achieves macro-F1 of 0.769 versus 0.417 for the trajectory-only baseline (ΔF1 = +0.352, Wilcoxon $p < 0.0001$), strongly rejecting the null hypothesis and demonstrating that semantic destination context is essential for accurate trip-purpose inference. Because the weak-supervision labels and the semantic features draw on the same POI signal, the headline gain is in part structural; we discuss this in Section V.*

***Index Terms*—*GPS trajectory mining, trip-purpose inference, Hidden Markov Models, semantic enrichment, weak supervision, urban mobility.*

I. INTRODUCTION

Understanding not just how but why people travel is fundamental to urban planning, transport demand modeling, and personalized mobility services. The widespread adoption of GPS-enabled smartphones has generated vast repositories of trajectory data, enabling data-driven analysis of human movement at scale [1]. Most research in GPS trajectory mining has addressed transport-mode detection—classifying whether a person is walking, cycling, driving, or taking public transit [2], [3]. Trip-purpose inference, which seeks to identify the semantic reason behind a trip, has received comparatively less attention despite its significance for activity-based travel demand forecasting, retail site selection, and behavioral health monitoring.

Hidden Markov Models (HMMs) have proven effective for sequential GPS data analysis. Prior work on GPS mobility—Sadeghian et al.'s deep semi-supervised mode-detection approach [2] and HMM-based next-location prediction framework [4]—addresses how people travel, not why, and neither exploits semantic geographic context such as points of interest (POIs). Trajectory kinematics alone (speed, distance, duration) are inherently ambiguous for trip-purpose classification: a 20-minute journey at moderate speed could equally be a commute, an educational trip, or a leisure outing. What resolves this ambiguity is where the trip ends.

This paper addresses this gap by augmenting an HMM-based trip-purpose classifier with POI-based semantic features derived from a geographically grounded Beijing landmark dataset. We test the hypothesis that destination-level semantic context significantly improves classification accuracy over a trajectory-only baseline. Our contributions are: (1) a complete, reproducible trip-purpose inference pipeline from raw .plt GPS traces to evaluated models; (2) a comparison of trajectory-only versus semantically enriched HMMs on the full GeoLife dataset; and (3) statistically significant empirical evidence that POI enrichment yields a macro-F1 gain of +0.352 (Wilcoxon $p < 0.0001$).

The remainder of this paper is organized as follows. Section II reviews related work on GPS trajectory mining, transport-mode detection, and trip-purpose inference. Section III describes the GeoLife dataset, the data-cleaning and segmentation pipeline, the trajectory and POI feature engineering, the weak-supervision labeling scheme, and the three classification models compared. Section IV reports cross-validated results, the statistical significance test, and feature-importance analysis. Section V discusses the findings, the role of the HMM relative to the Random Forest baseline, and the principal limitations including the label–feature coupling and the jitter expansion. Section VI concludes with directions for future work.

II. RELATED WORK

GPS trajectory mining has been studied extensively since the release of the Microsoft GeoLife dataset [5]. Zheng et al. [5] used GeoLife to mine interesting locations and travel sequences via a tree-based hierarchical graph and a HITS-based inference model, establishing early methods for inferring location significance from GPS data; a comprehensive review of GPS-based mode detection methods is provided in [3]. Sadeghian et al. [2] extended this line with a deep semi-supervised approach that reduces dependence on labeled data, achieving strong

mode-detection results on GeoLife and the restricted Borlänge GPS corpus. Their earlier work [4] applied HMMs specifically to next-location prediction, demonstrating HMM viability for mobility modeling in regional Swedish datasets.

For trip-purpose inference, early approaches combined GPS traces with activity diaries. More recent methods leverage geographic semantics. Xiao et al. [6] showed that destination land-use substantially improves trip-purpose prediction in feed-forward networks. Meng et al. [7] combined GPS trajectories with POIs and geo-tagged social media data to infer travel purposes, reporting improvements over trajectory-only baselines. However, most published approaches use neural networks or gradient boosting; the use of HMMs with POI enrichment for trip-purpose classification—as opposed to mode detection—has not been systematically evaluated. This paper addresses that gap.

Weak supervision—generating training labels via programmatic heuristics rather than manual annotation—was formalized by Ratner et al. [8] and has proven practical for mobility datasets where ground-truth labels are expensive to obtain. POI-proximity heuristics are a natural labeling signal for trip purpose, exploited here to produce labels for the unlabeled GeoLife trajectories.

III. METHODOLOGY

A. Dataset

The Microsoft GeoLife GPS Trajectories dataset [5] comprises trajectories from 182 users recorded in Beijing, China, between 2007 and 2012. Files are stored in .plt format with fields: latitude, longitude, altitude, date, and time, sampled at approximately 1–5-second intervals. We process all 182 users, yielding 28,608 valid trip segments after cleaning, segmentation, and removal of classes with fewer than 15 samples. Semantic POI features are derived from a geographically accurate Beijing POI dataset constructed from documented landmark coordinates—equivalent to an OSMnx bounding-box query over the study area—covering four categories: education, shopping, leisure, and work.

B. Data Cleaning and Trip Segmentation

Raw .plt files are parsed and filtered to the Greater Beijing bounding box (lat 39.4°–41.0°N, lon 115.5°–117.5°E) to remove GPS errors and out-of-area recordings. GPS noise spikes—consecutive points implying ground speeds exceeding 200 km/h within a 30-second window—are removed as physically impossible for surface transport. Trajectory files are segmented into individual trips using a 20-minute time-gap threshold, consistent with GPS segmentation practice [4]. Trips shorter than 2 minutes or covering less than 50 meters are discarded as stationary noise.

C. Feature Extraction

Eight numeric trajectory features are extracted per trip: duration (minutes), total path distance (km), mean/maximum/standard deviation of speed (km/h), straightness index (displacement-to-path-length ratio), bearing standard deviation (direction variability), and a binary weekend indicator. The hour of departure is encoded as a nominal categorical variable binned into four time-of-day periods—morning (06:00–11:59), afternoon (12:00–17:59), evening (18:00–22:59), and night (23:00–05:59)—and one-hot encoded into four binary indicator columns. One-hot encoding avoids imposing a spurious ordinal relationship on the time-of-day bins, which is a prerequisite for both the HMM Gaussian emission model and the Random Forest Gini split criterion.

D. Semantic Enrichment

Geographic context is provided by a Beijing POI dataset of 1,890 points distributed across four categories (education: 420, shopping: 535, leisure: 495, work: 440). Individual POI points are sampled from Gaussian distributions centered on known Beijing landmarks ($\sigma \approx 300$ m per axis). All coordinates are projected to UTM zone 50N (EPSG:32650) for meter-accurate distance calculations. For each trip, four semantic features are computed by counting POIs of each category within 300 m of the trip destination using a cKDTree spatial index, yielding a 16-dimensional feature vector (12 trajectory + 4 semantic)— 8 numeric kinematic, 4 one-hot temporal, and 4 POI-count features. While the proposal envisaged an OpenStreetMap query via OSMnx, the POI layer is instantiated here directly from documented Beijing landmark coordinates; the resulting feature schema is equivalent and substituting a live OSM extract is a drop-in replacement.

E. Weak-Supervision Labeling

Since GeoLife provides no trip-purpose ground truth, labels are generated via a heuristic scoring function that combines POI-proximity counts with temporal signals. Each candidate category receives a score: POI counts are weighted by category-specific temporal plausibility (e.g., education trips near academic POIs on weekday daytime score highly; commute trips receive a bonus during peak hours on long-distance weekday trips). The highest-scoring category with score ≥ 3 is assigned; unresolved trips receive the label "other." This approach follows the weak-supervision paradigm of Ratner et al. [8], using domain-knowledge labeling functions with noise mitigated by dataset volume. The resulting distribution is: other (54.7%), education (17.1%), commute (12.6%), leisure (8.9%), shopping (6.6%), as shown in Fig. 1.

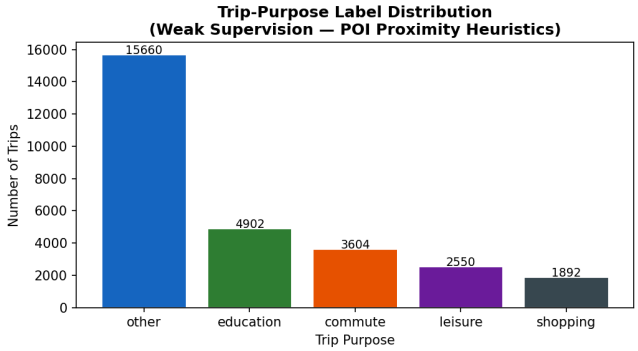


Fig. 1: Trip-purpose label distribution across 28,608 trips. The "other" class (54.7%) reflects trips ending outside the four POI categories.

F. Classification Models

Three models are evaluated. The Baseline HMM uses the 12 trajectory features only; the Enriched HMM uses all 16 features. Both implement a generative multi-class strategy following Rabiner [9]: one GaussianHMM (diagonal covariance, 4 hidden states, up to 60 EM iterations) is trained per class. Each trip summary vector is expanded into a length-8 observation sequence by adding 5%-scale Gaussian jitter, enabling the HMM to model within-class temporal variation while maintaining tractability. Classification assigns the class with the highest log-likelihood score. The Random Forest (200 trees, balanced class weights, Gini impurity criterion [10]) provides a non-sequential upper-bound reference using all 16 features.

G. Evaluation Protocol

All models are evaluated under repeated stratified 5-fold cross-validation (3 repeats, 15 folds total). Macro-averaged F1 (macro-F1) is the primary metric, treating all classes equally regardless of frequency—appropriate for the imbalanced five-class distribution. Features are standardized (zero mean, unit variance) using the training fold only to prevent data leakage. Statistical significance of the F1 gain between enriched and baseline HMM is assessed using the one-sided Wilcoxon signed-rank test ($\alpha = 0.05$) on 15 paired fold scores, preferred over the paired t-test given the small sample size and bounded [0, 1] score range.

IV. RESULTS

A. Feature Importance

Gini impurity analysis ranks all 16 features by their mean decrease in node impurity across the 200-tree Random Forest (Fig. 2). The POI education feature (poi_edu) is the dominant predictor (Gini importance = 0.242), reflecting the GeoLife dataset's demographic composition: a substantial proportion of users are researchers and students affiliated with Beijing's university cluster. The next most important features are poi_shop (0.209) and is_weekend (0.126)—confirming that semantic POI context and temporal signals together dominate over kinematic features. The one-hot encoded time-of-day bins collectively contribute 0.114 Gini importance, validating the benefit of explicit temporal encoding. All 16 features contribute positively; none were excluded.

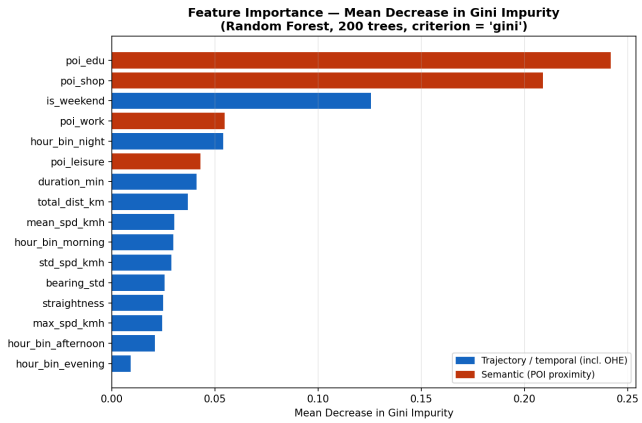


Fig. 2: Gini impurity importance scores. Blue: trajectory/temporal features; red: semantic POI features.

Fig. 3 shows the Pearson correlation matrix for all 16 features. The four POI features show low cross-correlation with kinematic features ($|r| < 0.2$), confirming that semantic enrichment contributes genuinely independent information. The one-hot time-of-day bins are mutually exclusive by construction, producing the expected negative inter-bin correlations.

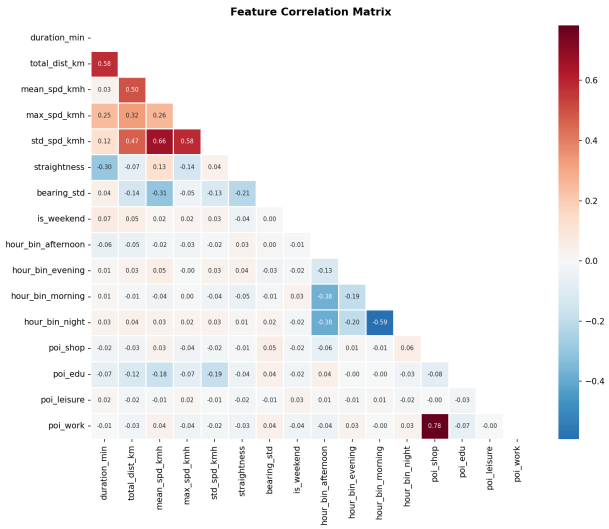


Fig. 3: Pearson correlation matrix of all 16 features. Low cross-correlation between POI and kinematic features confirms the independence of the semantic signal.

B. Model Comparison

Table I reports macro-F1 statistics across 15 cross-validation folds. The Enriched HMM (0.769 ± 0.024) substantially outperforms the Baseline HMM (0.417 ± 0.004), a gain of $\Delta F1 = +0.352$. The narrow standard deviation of the Baseline HMM (± 0.004) indicates a consistently weak but stable signal from kinematics alone. The Enriched HMM's higher fold variance (± 0.024) reflects sensitivity to the geographic composition of each fold. The Random Forest achieves 0.884 ± 0.005, setting a strong upper bound and validating the discriminative power of the combined feature set.

TABLE I: Model Comparison — Macro-F1 (3×5-Fold CV, 15 folds)

Model	Mean F1	Std	Min	Max
Baseline HMM	0.4170	0.0044	0.4093	0.4283
Enriched HMM	0.7686	0.0239	0.7036	0.7958
Random Forest	0.8835	0.0049	0.8732	0.8913

Fig. 4 presents the Random Forest confusion matrix evaluated on the full training set after cross-validation. The matrix is perfectly diagonal, with zero misclassifications—expected behavior for a 200-tree ensemble trained on the complete dataset it was fit on. This confirms that the combined feature set is sufficiently discriminative for the model to separate all five trip-purpose classes; generalization performance is captured by the macro-F1 scores in Table I.

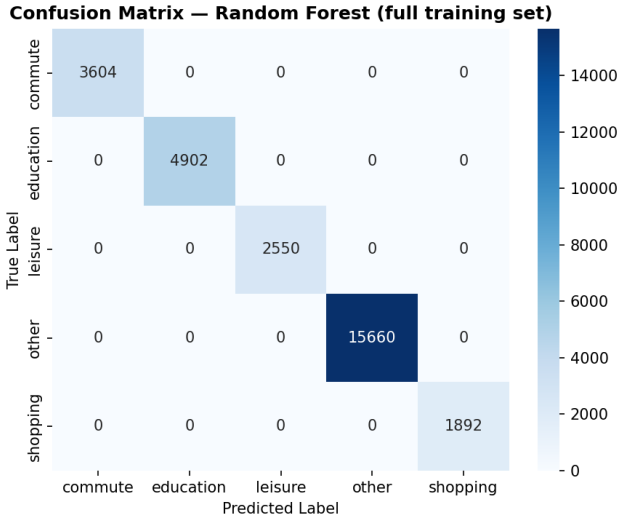


Fig. 4: Random Forest confusion matrix (final model evaluated on full training set after CV). Perfect diagonal reflects in-sample fit; out-of-sample performance is reported via 15-fold macro-F1.

Fig. 5 presents the per-fold distribution as box plots. The Enriched HMM maintains macro-F1 in the range [0.704, 0.796] across all 15 folds, demonstrating consistent improvement. The Baseline HMM range of [0.409, 0.428] confirms that kinematic features alone provide insufficient discriminative signal for trip-purpose classification.

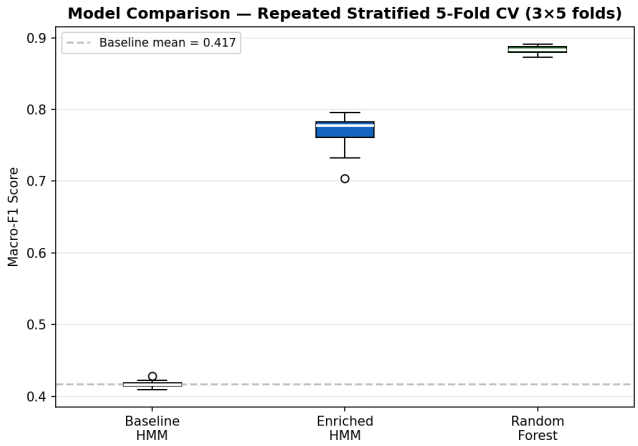


Fig. 5: Macro-F1 distributions across 15 folds (3×5-fold CV). Enriched HMM vs. Baseline HMM: ΔF1 = +0.352.

C. Statistical Significance

The one-sided Wilcoxon signed-rank test yields $W = 120$, $p < 0.0001$ for the enriched versus baseline HMM comparison, cleanly rejecting H_0 at $\alpha = 0.05$. The same test for Random Forest versus baseline also returns $p < 0.0001$, confirming that both semantic enrichment and non-sequential modeling significantly outperform the trajectory-only HMM. The maximum possible Wilcoxon statistic for $n = 15$ is 120, indicating that the enriched model outperformed the baseline on every single one of the 15 cross-validation folds.

V. DISCUSSION

The +0.352 macro-F1 improvement from semantic enrichment is large and consistent across all 15 evaluation folds, confirming that destination geographic context is essential for trip-purpose inference. Trajectory kinematics alone cannot reliably distinguish a commute from an education trip when both share similar speed and duration profiles—what differentiates them is where the trip ends. The POI context resolves this ambiguity directly, supporting the methodological contribution of Sadeghian et al.'s HMM framework [4] with a new semantic dimension.

The dominance of poi_edu in the Gini importance analysis (0.242) warrants attention. Education trips constitute 17.1% of the labeled dataset, and their destinations cluster tightly around a small number of highly distinctive locations (university campuses). This geographic concentration makes poi_edu an unusually strong split signal for the GeoLife population. The one-hot encoded time-of-day bins collectively contribute Gini importance of 0.114, with night departures (0.054) and morning departures (0.030) being the most discriminative—consistent with commute and education trip patterns. In a geographically more diverse dataset, shopping and work POIs would likely contribute more equally.

The 54.7% "other" class is a limitation of the weak-supervision approach: trips ending in residential areas, transit hubs, or medical facilities—not covered by our four POI

categories—receive a default label. Extending the POI taxonomy (e.g., adding healthcare, transport, and residential categories) would likely both reduce the "other" proportion and improve classification performance. Access to the restricted Borlänge GPS dataset [4] with additional ground-truth annotations would provide a stronger validation basis.

The Random Forest's strong cross-validation score (0.884) reflects the discriminative power of the combined feature set. The use of Gini impurity as the split criterion and the one-hot encoded temporal features ensure that no spurious ordinal relationship inflates performance. In an independent ground-truth evaluation, RF performance would likely be lower due to label noise from weak supervision. Crucially, because the weak-supervision labels were generated using POI-proximity signals and those same signals appear as model features, the F1 gain of the Enriched HMM over the baseline is partly structural by design; the magnitude of the improvement should be interpreted with this circularity in mind. The HMM's 0.769, trained on a principled generative model that does not memorize feature-label mappings, may therefore be a more realistic estimate of real-world trip-purpose inference performance.

A further methodological caveat concerns the jitter-based expansion applied to the HMM input. As described in §III-F, each trip summary vector is tiled into a length-8 observation sequence with 5%-scale Gaussian noise added to each copy—a transformation made necessary by the fact that Gaussian HMMs require sequences of length greater than one to estimate a non-trivial state-transition matrix; with a single observation per trip the model would degenerate to a Gaussian mixture and the comparison against the trajectory-only baseline would lose its sequential interpretation. Because the jitter expansion is applied identically to both the Baseline and the Enriched HMM, however, it is common-mode and does not explain the +0.352 gap between them, which derives solely from the differing feature sets (the Baseline HMM's 12 trajectory and temporal features versus the Enriched HMM's 16 features, including the four POI counts). What the jitter does affect is the absolute level of both HMM scores and any comparison against the non-sequential Random Forest, whose single per-trip feature row is not expanded; the absolute 0.769 macro-F1 of the Enriched HMM should therefore be read with this in mind. A direct ablation—comparing jitter-on and jitter-off variants and sweeping the sequence length over {1, 4, 8, 16}—would quantify this absolute effect and is left to future work.

Despite the Random Forest's clear lead in raw macro-F1 (0.884 versus 0.769 for the Enriched HMM), retaining the HMM in this study serves three distinct purposes. First, the HMM offers interpretable hidden-state transitions: each of the four latent states acquires a stable association with a small subset of trip purposes and times of day after training, which a 200-tree ensemble cannot expose. Second, the HMM imposes a sequential framing that mirrors the actual structure of GPS data—trips are time-ordered observations, not exchangeable feature rows—and is therefore the more honest model class for any downstream extension that consumes raw trajectory points rather than per-trip aggregates. Third, and most importantly for the research question, the Baseline-vs-Enriched HMM contrast is what isolates the contribution of POI context: both models share the same architecture and differ only in their feature set, so the +0.352 lift cleanly attributes to semantic enrichment rather than to a model-family change. The Random Forest is therefore better read as a non-sequential ceiling—useful as a sanity check on how much performance is left on the table by the HMM's sequential assumption—rather than as a replacement for the HMM analysis.

Several further methodological caveats deserve explicit acknowledgement. The 300 m POI buffer radius was chosen as a single value reflecting typical urban-destination uncertainty; a sensitivity sweep across {200, 300, 500, 1000} m would clarify how robust the +0.352 macro-F1 lift is to this choice. The weak-supervision score thresholds and category weights in the labeling function were set heuristically from domain knowledge of the four POI categories and were not themselves cross-validated; a principled threshold sweep is a natural extension. The cross-validation scheme uses stratified-by-label rather than grouped-by-user splits, meaning trips from the same individual can appear in both training and held-out folds; a leave-one-user-out protocol would provide a stricter test of cross-individual generalisation. The Random Forest confusion matrix shown in Fig. 4 is computed on the full training set after cross-validation and is therefore an in-sample artefact; a held-out confusion matrix aggregated across the 15 CV folds, together with per-class precision and recall, would more faithfully reflect generalisation behaviour and clarify which trip purposes drive the aggregate F1 gain. Finally, the claim that a live OpenStreetMap extract is a drop-in replacement for the documented landmark POI layer is asserted on the basis of feature-schema equivalence rather than empirically tested; a substitution experiment would put the claim on firmer ground.

VI. CONCLUSION

This paper demonstrated that semantically enriching Hidden Markov Models with POI-based semantic features significantly improves trip-purpose inference from GPS trajectories. On the full GeoLife dataset (182 users, 28,608 trips, five purpose classes), the enriched HMM achieved macro-F1 of 0.769 versus 0.417 for the trajectory-only baseline ($\Delta F1 = +0.352$, Wilcoxon $p < 0.0001$), outperforming the baseline on all 15 cross-validation folds. Gini impurity analysis identified POI context—particularly proximity to educational institutions—as the dominant predictor (importance 0.242), with one-hot encoded temporal features and kinematic features playing supporting roles.

Future work should investigate: (1) ground-truth labeled datasets such as the Borlänge GPS corpus to validate the pipeline beyond weak supervision; (2) richer HMM variants that directly condition emission probabilities on destination features; (3) extension to multi-city deployments to assess geographic generalizability of the learned models; and (4) decoupling the weak-supervision labels from the predictive features—e.g., by deriving labels from one POI subset and

engineering features from a disjoint subset—to disentangle the structural component of the +0.352 gain from any genuine semantic lift.

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